Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software Engineer graduating May 2026 with 2+ years building Python services, full-stack applications, and cloud data platforms using FastAPI, Flask, React, AWS, PostgreSQL, Docker, CI/CD, and automated testing.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, California*

- Built ETL-style integrations across FastAPI, SQL Server, Supabase, and cloud storage, processing 5K+ monthly jobs with reconciliation checks, error handling, and CI/CD-backed releases achieving 99%+ production reliability.
- Designed a multi-tenant PostgreSQL schema supporting 200+ users and 10K+ structured records, implementing indexing, query tuning, validation, and row-level isolation that reduced production latency by 35% consistently.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, California*

- Implemented a production Flask backend on AWS serving 150+ instructors and 10K+ grading requests daily, with secure ingestion, automated tests, CI/CD deployment, and stable API performance under peak usage.
- Developed React interfaces and backend workflows for uploads, grading history, rubric generation, and concurrent evaluation, reducing feedback turnaround by 60% while maintaining reliable user-facing service behavior at scale.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, California*

- Built a Python model-evaluation pipeline processing 100K+ trajectory logs, improving long-horizon planning 18% through reproducible data transformations, benchmark automation, regression testing, and documented experiment results across releases.
- Developed a spatial-temporal encoder combining map geometry and agent context, improving closed-loop stability 15% across complex scenarios through systematic failure analysis, controlled evaluation, and root-cause debugging in Python.

Aasma Technology Solutions **August 2023 – July 2024**
Software Development Engineer, SecureSync *India*

- Engineered cross-platform microservices with automated VM builds and 20+ unit and integration suites using Jenkins, PyTest, and Mockito across Windows, Linux, and macOS production release environments consistently.
- Troubleshot AWS Glue, S3, and API integrations, improving response times 30% and pipeline efficiency 68% while reducing technical debt 20% through reusable services, validation, and code reviews.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern, VibrantChem *India*

- Built a MERN resource-management platform with REST APIs, WebSockets, and microservices, supporting six operational workflows and 2M+ synchronized records while reducing code-review turnaround by 25% across releases.
- Automated application deployments with Jenkins CI/CD and Docker packaging, achieving 95%+ successful build validation and cutting cloud release time by 50% through repeatable environment configuration across deployments.

Skills

Languages: Python, JavaScript, TypeScript, Java, C++, SQL, Bash
Backend & Frontend: FastAPI, Flask, Django, Django REST Framework, React, Vue.js, REST APIs, WebSockets, Microservices, OOP
Cloud & Data: AWS, S3, RDS, PostgreSQL, SQL Server, Azure SQL, Supabase, Docker, Kubernetes, ETL, Data Validation
Quality & Delivery: CI/CD, Jenkins, GitHub Actions, PyTest, Unit Testing, Integration Testing, Debugging, Monitoring, Agile/Scrum, Git

Projects

HMIS - Hospital Management Information System Live | *TypeScript, ABDM APIs, Twilio, Sentry*

- Implemented consent-based patient-history access, appointment workflows, alerts, and production monitoring for a healthcare platform, reducing administrative workload 30% while supporting secure clinician-facing access to structured records in production.

Big-Brain Live | *LLaMA-2, FAISS, PostgreSQL, GCP*

- Engineered a cloud ingestion and indexing pipeline processing 50K+ records, implementing incremental updates, metadata persistence, validation, and query tuning that reduced retrieval latency by 40% for document workloads.

Linguistix | *Python, AWS, Docker, CI/CD*

- Developed and deployed a containerized Python REST API on AWS using CodePipeline and ECS Fargate, with automated releases, validation checks, fault handling, and dependable recurring data-processing workflows.